

Assignment Day – 3

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.



No.	BIOS/POST Error Message	Usually Indicates
1	CMOS Checksum Error	CMOS battery is weak/dead or BIOS settings are corrupted.
2	Keyboard Error or No Keyboard Present	Keyboard is disconnected or the keyboard/USB port has a problem.
3	Memory Size Error / Memory Error	RAM is not properly installed, faulty, or incompatible.
4	CPU Fan Error	CPU fan is not connected, not spinning, or has a cooling problem.
5	No Boot Device Found	HDD/SSD is not detected, or the boot device/order is configured incorrectly.

2. **Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.**



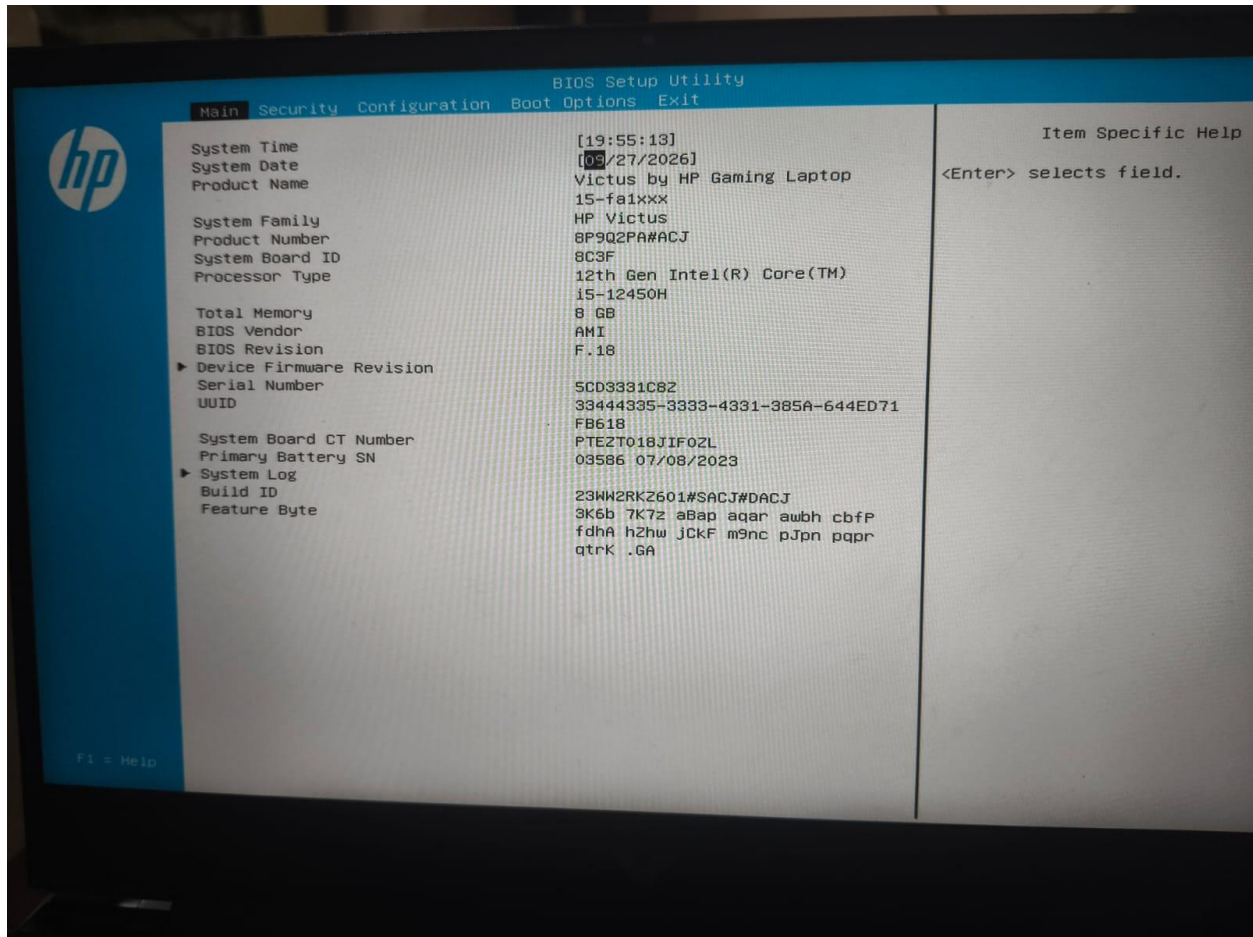
For an AMI BIOS, 3 short repeating beeps typically indicate a base 64 KB memory (RAM) failure. This means the BIOS is unable to properly read/write to the system memory during POST.

Two troubleshooting steps:

1. **Power off the PC and reseal the RAM — remove the RAM module and install it again firmly in the correct slot.**
2. **Test the RAM modules one at a time — if there are multiple sticks, test each separately or try a known-good compatible RAM module.**

Note: Beep codes can vary by motherboard/firmware, so the motherboard's manual should be checked if the code doesn't match.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



- 4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.**

Hint: Think about the CMOS battery and BIOS settings reset.



CMOS checksum error – Defaults loaded means the BIOS has detected that the stored CMOS/BIOS settings are invalid or corrupted, so it has loaded the default settings.

How to resolve it:

- 1. Enter the BIOS and set the correct date, time, boot order, and other required settings, then save and restart.**
- 2. If the error appears again, replace the CMOS/RTC battery because it may be weak or dead.**
- 3. If necessary, clear/reset the CMOS using the motherboard's recommended procedure, then configure the BIOS settings again.**